

Differential Equation

5. Exact Differential Equation

If the equation is of the form $M(x, y)dx + N(x, y)dy = 0$

And it follows, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Then the solution may be written as

$$u = \int M dx + k(y)$$

$$u = \int N dy + l(x)$$

Comparing both the terms the solution may be written.

Example: An exact equation

$$(x^3 + 3xy^2)dx + (3x^2y + y^2)dy = 0.$$

Solution: Step1. Test for exactness.

$$M = x^3 + 3xy^2, \quad N = 3x^2y + y^2$$

$$\frac{\partial M}{\partial y} = 6xy, \quad \frac{\partial N}{\partial x} = 6xy$$

It is exact.

Step 2: Implicit solution

$$u = \int M dx + k(y) = \int (x^3 + 3xy^2) dx + k(y) = \frac{1}{4}x^4 + \frac{3}{2}x^2y^2 + k(y).$$

To find $k(y)$, we differentiate this formula with respect to y

$$\frac{\partial u}{\partial y} = 3x^2y + \frac{dk}{dy} = N = 3x^2y + y^3$$

Hence $dk / dy = y^3$ so that $k = (y^4 / 4) + c$.

$$u(x, y) = \frac{1}{4}(x^4 + 6x^2y^2 + y^4) = c.$$

Step 3: Checking. Note well that the present method gives the solution in implicit form,

$u(x, y) = c = \text{const}$, not in explicit form, $y = f(x)$.

For checking, we can differentiate $u(x, y) = c$ implicitly and see whether this leads to

$\frac{dy}{dx} = -\frac{M}{N}$, the given equation. In the present case, differentiating (9) implicitly with respect to

x , we obtain

$$\frac{1}{4}(4x^3 + 12xy^2 + 12x^2yy' + 4y^3y') = 0.$$

Collecting terms, we see that this equal $M + Ny' = 0$; thus $M dx + N dy = 0$. This completes the check.

Example: An initial value problem

Solve the initial value problem

$$(\sin x \cosh y) dx - (\cos x \sinh y) dy = 0, \quad y(0) = 3.$$

Solution: we can verify that this equation is exact.

$$u = \int \sin x \cosh y dx + k(y) = -\cos x \cosh y + k(y).$$

From this, $\partial u / \partial y = -\cos x \sinh y + dk / dy$. Hence $dk / dy = \cos x \sinh y$, and $k = \cosh y \cos x$. The general solution is $u = \text{const}$, that is, $\cos x \cosh y = c$.

The initial condition gives $\cos 0 \cosh 3 = 10.07 = c$. Hence the answer is $\cos x \cosh y = 10.07$.

Example: WARNING! Breakdown in the case of non-exactness

Consider the equation

$$-y dx + x dy = 0.$$

We see that $M = -y, N = x$, hence $\partial M / \partial y = -1$ but $\partial N / \partial x = 1$. Hence the equation is not exact in such a case, the present method does not work.

Integrating Factor $F(x)$

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x), \text{ The I.F. } F(x) = e^{\int f(x) dx}$$

Integrating Factor $F(y)$

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = f(y), \text{ The I.F. } F(y) = e^{\int f(y) dy}$$

Example: Find an integrating factor and solve the initial value problem

$$2 \sin(y^2) dx + xy \cos(y^2) dy = 0, \quad y(2) = \sqrt{\pi/2}.$$

Solution: Step 1. Check for exactness. We have

$$P = 2 \sin(y^2) \text{ and } Q = xy \cos(y^2).$$

The equation is not exact because

$$P_y = 4y \cos(y^2) \neq Q_x = y \cos(y^2).$$

Step 2: Integrating factor

$$R = \frac{1}{Q}(P_y - Q_x) = \frac{1}{xy \cos(y^2)} [4y \cos(y^2) - y \cos(y^2)] = \frac{3y}{xy} = \frac{3}{x}.$$

$$F(x) = \exp \int R(x) dx = \exp \int \frac{3}{x} dx = x^3.$$

Multiplying the given equation by x^3 , we get the new equation

$$2x^3 \sin(y^2) dx + x^4 y \cos(y^2) dy = 0$$

This equation is exact because

$$\frac{\partial}{\partial y} [2x^3 \sin(y^2)] = 4x^3 y \cos(y^2) = \frac{\partial}{\partial x} [x^4 y \cos(y^2)].$$

Step 3: General solution.

$$u = \int 2x^3 \sin(y^2) dx = \frac{1}{2} x^4 \sin(y^2) + k(y).$$

$$u_y = x^4 y \cos(y^2) + k'(y) = x^4 y \cos(y^2).$$

Hence $k'(y) = 0$ and $k = \text{const.}$ This gives the general solution

$$u(x, y) = \frac{1}{2} x^4 \sin(y^2) = c = \text{const.}$$

Step 4: Particular solution of the initial value problem. Substituting the initial condition

$y(2) = \sqrt{\pi/2}$ into $u(x, y)$; we have

$$\frac{1}{2} \cdot 2^4 \sin \frac{\pi}{2} = 8 = c.$$

Hence the desired particular solution is

$$\frac{1}{2} x^4 \sin(y^2) = 8 \quad \text{or} \quad x^4 \sin(y^2) = 16.$$